

University of Toronto Mississauga

Applied Time Series Analysis

STA457H5S LEC9101

Winter 2022

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Midterm Solutions

Duration: 90 Minutes

February 28, 2022

INSTRUCTIONS and POLICIES:

- This exam contains 3 questions. Total points: 60 = 15 percent of your final mark.
- Answer ALL questions to get full credit. Do not skip steps in your solutions or fail to describe what you are doing.
- Complete solutions are required for all questions.
- Carry out all your computations up to 3 decimals.
- **20 additional minutes** will be given at the end to upload your solutions on Crowdmark.

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3. Posting test, essay, or exam questions to message boards or social media.
4. Creating, accessing, and sharing assessment questions and answers in virtual “course groups.”
5. Working collaboratively, in-person or online, with others on assessments that are expected to be completed individually.

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Question 1 (20 points). Suppose the stochastic processes $\{X_t\}$ and $\{Y_t\}$ are given by

$$X_t = -2t + W_t + 0.5W_{t-1}$$

and

$$Y_t = \nabla X_t = (1 - B)X_t,$$

respectively, where $\{W_t\} \stackrel{iid}{\sim} N(0, \sigma_w^2)$.

Find the *mean*, *variance*, *autocovariance function* and *autocorrelation function* of Y_t . Then, discuss whether $\{Y_t\}$ is a weakly stationary process by explicitly writing down the required conditions.

Solution.

$$\begin{aligned} Y_t &= (1 - B)X_t = X_t - BX_t \\ &= X_t - X_{t-1} \\ &= -2t + W_t + 0.5W_{t-1} - [-2(t-1) + W_{t-1} + 0.5W_{t-1-1}] \\ &= W_t - 0.5W_{t-1} - 0.5W_{t-2} - 2 \end{aligned}$$

$$E(Y_t) = E(W_t) - 0.5E(W_{t-1}) - 0.5E(W_{t-2}) - 2 = \boxed{-2}$$

$$\text{Var}(Y_t) = \text{Var}(W_t) + 0.5^2 \text{Var}(W_{t-1}) + 0.5^2 \text{Var}(W_{t-2}) = \boxed{1.5\sigma_w^2}$$

The autocovariance function can be expressed as follows:

$$\begin{aligned} \gamma_{t,t-h} &= E[(Y_t - E[Y_t])(Y_{t-h} - E[Y_{t-h}])] \\ &= E[(Y_t + 2)(Y_{t-h} + 2)] \\ &= E[(W_t - 0.5W_{t-1} - 0.5W_{t-2})(W_{t-h} - 0.5W_{t-h-1} - 0.5W_{t-h-2})] \end{aligned}$$

Then, by noting that

$$E[w_t w_s] = \begin{cases} \sigma_w^2 & \text{if } t = s \\ 0 & \text{if } t \neq s \end{cases}$$

and

$$E[W_t^2] = \text{Var}[W_t] - (E[W_t])^2 = \sigma_w^2 - 0 = \sigma_w^2,$$

we derive:

$$\begin{aligned}\gamma_{t,t-1} &= E[(W_t - 0.5W_{t-1} - 0.5W_{t-2})(W_{t-1} - 0.5W_{t-2} - 0.5W_{t-3})] \\ &= -0.5E[W_{t-1}^2] + 0.25E[W_{t-2}] = \boxed{-0.25\sigma_w^2}\end{aligned}$$

and

$$\begin{aligned}\gamma_{t,t-2} &= E[(W_t - 0.5W_{t-1} - 0.5W_{t-2})(W_{t-2} - 0.5W_{t-3} - 0.5W_{t-4})] \\ &= -0.5E[W_{t-2}^2] = \boxed{-0.5\sigma_w^2},\end{aligned}$$

whereas for lag values $h \geq 3$ we get:

$$\boxed{\gamma_{t,t-h} = 0}$$

Therefore, the final results can be summarized as follows:

$$\gamma_{t,t-h} = \begin{cases} 1.5\sigma_w^2 & \text{if } h = 0 \\ -0.25\sigma_w^2 & \text{if } h = 1 \\ -0.5\sigma_w^2 & \text{if } h = 2 \\ 0 & \text{if } h \geq 3 \end{cases}$$

The autocorrelation function is further derived as:

$$\rho_{t,t-h} = \frac{\gamma_{t,t-h}}{\gamma_{t,t}} = \frac{\gamma_{t,t-h}}{1.5\sigma_w^2}$$

$$\rho_{t,t-h} = \begin{cases} 1 & \text{if } h = 0 \\ -\frac{1}{6} & \text{if } h = 1 \\ -\frac{1}{3} & \text{if } h = 2 \\ 0 & \text{if } h \geq 3 \end{cases}$$

Hence, the process is stationary since the mean is constant, variance is finite (since it is a constant) and the autocovariance (or alternatively the autocorrelation) function is time-independent.

Question 2 (20 points). Consider the following mixed ARMA(p,q) model

$$X_t = -0.3X_{t-1} + 0.1X_{t-2} + W_t + 0.9W_{t-1} + 0.2W_{t-2},$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$.

- a) Determine the orders p and q of the given model (watch out for parameter redundancy).
- b) Is this model invertible? Briefly discuss the importance of invertibility by mentioning two analytical/practical features of invertible time series models.
- c) Show that the given model is causal and compute its corresponding ψ -coefficients in the following causal linear process representation

$$X_t = \sum_{j=0}^{\infty} \psi_j W_{t-j}.$$

- d) Using the fact that W_t is a random variable representing the *shock* (sudden pulse or news) exerted onto the system at time t , provide an *economic interpretation* (an interpretation based on the calculus of variations) of individual ψ -coefficients. After that, discuss what characteristics an exponential decay of the ψ -weights reveals about the past shocks.

Solution. Let us re-write the model as:

$$\begin{aligned} X_t + 0.3X_{t-1} - 0.1X_{t-2} &= W_t + 0.9W_{t-1} + 0.2W_{t-2} \\ \Rightarrow (1 + 0.3B - 0.1B^2) X_t &= (1 + 0.9B + 0.2B^2) W_t \end{aligned}$$

Part a) The MA characteristic polynomial for the given model is

$$\theta(B) = 1 + 0.9B + 0.2B^2$$

whose roots are found as being:

$$1 + 0.9B + 0.2B^2 = 0 \Rightarrow B = \frac{-0.9 \pm \sqrt{0.9^2 - 4(0.2)}}{2(0.2)} = -2 \text{ and } -2.5$$

Accordingly, we can factorize the $\theta(B)$ as:

$$\boxed{\theta(B) = 0.2 (B + 2)(B + 2.5)}$$

Furthermore, the AR characteristic polynomial is obtained as follows:

$$\phi(B) = 1 + 0.3B - 0.1B^2$$

Thus,

$$1 + 0.3B - 0.1B^2 = 0 \Rightarrow B = \frac{-0.3 \pm \sqrt{0.9 + 0.4}}{0.2} = -2 \text{ and } 5$$

which, in turn, yields

$$\boxed{\phi(B) = -0.1 (B + 2)(B - 5)}$$

It is seen that two $(B + 2)$ is a common factor which needs to be eliminated. It is only after removing this common factor that one **is allowed** to represent the given time series using the form

$$\phi(B)X_t = \theta(B)W_t.$$

In other words,

$$(B - 5)X_t = (B + 2.5)W_t \Rightarrow X_{t-1} - 5X_t = W_{t-1} + 2.5W_t$$

which implies that the model is ARMA(1,1), hence $p = 1$ and $q = 1$.

Part b) Yes, the model is invertible since the root of the reduced MA characteristic polynomial is $B = -2.5$ whose magnitude or modulus is strictly greater than one.

The invertibility ensures unique identifiable model is chosen. Furthermore, it enables one to rewrite the process as AR(∞).

Part c) The model is causal since the root of the reduced AR characteristic polynomial is $B = 5$, hence $|B| > 1$.

Part d) The coefficient measures how sensitive the process is with respect to the shock excreted at time t . The exponential decay of the weights shows that the process is insensitive mainly with respect to past shocks excreted to the system in near past.

Question 3 (20 points). Let $\{X_t\}$ be the process defined by

$$X_t - \mu = \phi(X_{t-1} - \mu) + W_t$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$, and the model parameters are given by $\phi = 0.6$ and $\sigma_w^2 = 2$. Suppose that in a sample of size 100 from this process we obtain $\bar{x}_{100} = 0.271$. First, construct an approximate 95% confidence interval for μ , and then investigate whether the data are compatible with the hypothesis that $\mu = 0$?

Solution. Since $\{X_t\}$ is an AR(1) process, we have that

$$\gamma_h = \frac{\phi^{|h|} \sigma_w^2}{1 - \phi^2}$$

We estimate μ by $\bar{X}_n$ which is approximately normally distributed with mean μ and variance $\frac{1}{n} \sum_{|h| < \infty} \gamma_h$. More precisely,

$$\begin{aligned} \frac{1}{n} \sum_{|h| < \infty} \gamma_h &= \frac{1}{n} \left(1 + 2 \sum_{h=1}^{\infty} \phi^h \right) \frac{\sigma^2}{1 - \phi^2} \\ &= \frac{1}{n} \left(1 + 2 \left(\frac{1}{1 - \phi} - 1 \right) \right) \frac{\sigma^2}{1 - \phi^2} \\ &= \frac{1}{n} \left(\frac{2}{1 - \phi} - 1 \right) \frac{\sigma^2}{1 - \phi^2} \\ &= \frac{1}{n} \left(\frac{1 + \phi}{1 - \phi} \right) \frac{\sigma^2}{1 - \phi^2} = \boxed{\frac{\sigma^2}{n(1 - \phi)^2}} \end{aligned}$$

Thus,

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n(1 - \phi)^2}\right)$$

A 95% confidence interval is then given by

$$\left(\bar{x}_n - z_{0.025} \frac{\sigma}{\sqrt{n}(1 - \phi)}, \bar{x}_n + z_{0.025} \frac{\sigma}{\sqrt{n}(1 - \phi)} \right) = (-0.419, 0.961) = 0.271 \pm 0.69$$

Since 0 belongs to the confidence interval, the hypothesis that $\mu = 0$ cannot be rejected.

Standard Normal Cdf

If $Z \sim N(0, 1)$, then we can use Table D.2 to compute the cumulative distribution function (cdf) Φ for Z . For example, suppose we want to compute $\Phi(z) = P(Z < 1.03)$. The symmetry of the $N(0, 1)$ distribution about 0 implies that $\Phi(z) = 1 - \Phi(-z)$, so using Table D.2, we have that $P(Z < 1.03) = P(Z < 1.03) = 1 - P(Z < -1.03) = 1 - 0.1515 = 0.8485$.

Table D.2 Standard Normal Cdf										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641